
Statistics for AI & ML

Hypothesis Testing, Correlation, CLT & Bayes' Theorem

What's inside

- Hypothesis Testing — t-test, Chi-square test, ANOVA
- Correlation vs Causation
- Central Limit Theorem & Sampling
- Bayes' Theorem Basics

Each topic pairs a plain-language explanation with a real-data worked example and its matching AI/ML use case.

A Study-cum-Teaching Module

Companion Jupyter notebook: `stats_for_aiml.ipynb`

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How to use this module

This handbook covers four statistics topics that show up constantly in AI/ML work: testing whether a difference is real (hypothesis testing), telling apart “related” from “causes” (correlation vs causation), understanding why sample averages behave predictably (Central Limit Theorem), and updating beliefs with evidence (Bayes’ theorem).

Every topic follows the same simple flow:

1. **Key Idea** — the concept in one or two lines.
2. **Intuition** — a plain-English mental picture, no formulas yet.
3. **Real-life example** — a everyday situation you already understand.
4. **Worked example** — the same idea, done with real/realistic data and numbers.
5. **AI/ML use case** — where this exact idea shows up in machine learning practice.
6. **Try it yourself** — a small exercise using the companion notebook.

NOTE

The companion notebook `stats_for_aiml.ipynb` has the full working code for every example in this handbook, in the same order. Run it side by side with this document.

1 Hypothesis Testing

★ KEY IDEA

Hypothesis testing is a formal way of asking: “*Is this difference/pattern I’m seeing in my sample real, or could it just be random noise?*” We start by assuming nothing interesting is happening (the **null hypothesis**, H_0), then check if the data gives us strong enough evidence to reject that assumption.

INTUITION

Think of it like a courtroom. The default assumption is “**innocent**” (**no effect, no difference**) — this is H_0 . The prosecution (your data) has to bring enough evidence to convince the jury *beyond reasonable doubt*. The “reasonable doubt” threshold is the **significance level**, usually $\alpha = 0.05$. If the evidence is strong enough ($p < 0.05$), we reject H_0 and say the effect is **statistically significant**.

NOTE

The **p-value** is the probability of seeing a result at least this extreme *if H_0 were actually true*. A small p-value means “this would be a very unlikely coincidence if there were truly no effect” — so we doubt H_0 .

We’ll look at three common tests, each suited to a different kind of question.

1.1 The t-test — comparing two group means

INTUITION

Use a t-test when you want to know: “*Do these two groups have different averages?*” For example: do smokers tip differently than non-smokers at a restaurant?

WORKED EXAMPLE

Data: The classic `tips` dataset (244 real restaurant bills, tip amounts, and customer attributes such as smoker status).

Question: Do smokers and non-smokers leave different average tips?

Result (independent two-sample t-test):

Group	Mean tip (\$)	n
Smoker	3.01	93
Non-smoker	2.99	151

$t = 0.09$, $p = 0.93$.

Interpretation: $p = 0.93 \gg 0.05$, so we **fail to reject** H_0 . The tiny \$0.02 gap is well within random variation — there’s no real evidence that smokers tip differently.

AI/ML USE CASE

Comparing two ML models’ accuracy across cross-validation folds, or checking if a new feature/model version genuinely improves a metric (A/B testing a recommendation algorithm’s click-through rate between version A and version B).

1.2 The Chi-square test — are two categories related?

INTUITION

Use a chi-square test when both variables are **categories**, not numbers — e.g. “passenger class” and “survived or not”. It compares what you *actually observed* in each combination of categories against what you’d *expect* if the two variables were totally unrelated.

WORKED EXAMPLE

Data: The Titanic passenger dataset (891 real passenger records).

Question: Is survival related to passenger class (1st/2nd/3rd)?

Observed survival rate by class:

Class	1st	2nd	3rd
Survival rate	63%	47%	24%

$\chi^2 = 102.9$, $df = 2$, $p < 0.001$.

Interpretation: p is extremely small, so we **reject** H_0 . Class and survival are clearly related — 1st class passengers survived at more than double the rate of 3rd class passengers.

AI/ML USE CASE

Checking whether a categorical feature (e.g. `region`, `device_type`) is actually associated with the target label before including it in a model, or testing independence between a model’s predicted class and a sensitive attribute (a basic fairness check).

1.3 ANOVA — comparing more than two group means

INTUITION

A t-test compares 2 groups. What if you have 3 or more? Running many t-tests pairwise inflates your false-positive risk. **ANOVA (Analysis of Variance)** compares all groups at once and answers: “*Is at least one group’s average different from the rest?*”

WORKED EXAMPLE

Data: The same `tips` dataset, grouped by day of the week (Thu, Fri, Sat, Sun).

Question: Does the average total bill differ across days?

Day	Thu	Fri	Sat	Sun
Mean bill (\$)	17.68	17.15	20.44	21.41

$F = 2.77$, $p = 0.042$.

Interpretation: $p = 0.042 < 0.05$, so we **reject** H_0 — average bill size is not the same across all four days (weekend bills run noticeably higher).

AI/ML USE CASE

Comparing model performance (e.g. RMSE) across *three or more* algorithms (Linear Regression vs Random Forest vs XGBoost) trained on the same data, or comparing user engagement across more than two app-design variants in a multi-arm A/B test.

WATCH OUT

A significant p-value tells you *something* is different – it does not tell you *how big* the difference is, or which specific groups differ (for ANOVA, you’d follow up with a post-hoc test like Tukey’s HSD). Also, “statistically significant” is not the same as “practically important” — with a huge sample size, even a tiny, meaningless difference can produce a small p-value.

TRY IT YOURSELF

Open the notebook and pick a different pair of columns from the `tips` dataset (e.g. lunch vs dinner). Re-run the t-test. Is the difference in tips significant? Then try grouping by `size` (party size) and run an ANOVA on total bill.

2 Correlation vs Causation

★ KEY IDEA

Correlation means two variables move together. **Causation** means one variable directly *makes* the other change. Two things being correlated **does not prove** that one causes the other.

INTUITION

Correlation only tells you two lines on a graph move in a similar pattern. It says nothing about *why*. Often, a hidden third variable — a **confounder** — is driving both.

WORKED EXAMPLE

Real-life example: Every summer, both ice-cream sales and drowning incidents rise together.

Month	Avg. temp (°C)	Ice-cream sales (\$k)	Drowning incidents
Jan	12	8	2
Apr	22	25	6
Jul	34	58	15
Oct	20	22	5

Correlation between ice-cream sales and drownings: $r = 1.00$ — almost perfect.

Question: Does eating ice-cream cause drowning?

Answer: No. The **confounding variable** is **temperature**. Hot weather makes more people buy ice-cream *and* makes more people go swimming (which raises drowning risk). Ice-cream and drowning are correlated only because they share a common cause.

NOTE

Correlation is measured by the **Pearson correlation coefficient** r , ranging from -1 (perfect negative relationship) to $+1$ (perfect positive relationship), with 0 meaning no linear relationship. r alone never tells you about direction of cause.

WORKED EXAMPLE

Data check on real data: In the `tips` dataset, `total_bill` and `tip` are correlated at $r = 0.68$ — bigger bills come with bigger tips, which makes intuitive sense (tips are usually a percentage of the bill, so this is plausibly causal). Compare that to the ice-cream/drowning case, where the relationship is real but the story behind it is a confounder, not cause.

AI/ML USE CASE

Feature selection traps: a feature can be highly correlated with your target label purely through a confounder or **data leakage** (e.g. a hospital-readmission model finding that “number of tests ordered” predicts illness severity — but doctors order more tests *because* the patient is already sicker, not the other way around). Models trained on such correlations can look great on paper and fail in the real world once the confounding relationship shifts. This is exactly why causal inference and A/B testing (randomized experiments) matter — randomization is what lets you claim causation, not just correlation.

WATCH OUT

High correlation between two *features* you plan to feed into a model (**multicollinearity**) can also destabilize linear models’ coefficients, even with no causal question involved at all.

TRY IT YOURSELF

In the notebook, compute the correlation between `party_size` and `tip` in the `tips` dataset. Is it strong? Can you think of a confounder that might explain part of the relationship (hint: bigger parties → bigger bills → bigger tips)?

3 Central Limit Theorem & Sampling

★ KEY IDEA

The **Central Limit Theorem (CLT)** says: if you repeatedly take random samples from *any* population and compute the **sample mean** each time, those sample means will form a roughly **normal (bell-shaped) distribution** — even if the original population itself is skewed or weirdly shaped — as long as the sample size is reasonably large (commonly $n \geq 30$).

INTUITION

Imagine India's household income — a handful of very high earners create a long right-skewed tail; most people earn far less than the mean. If you took just *one* household's income as your estimate, you'd get wildly different numbers each time. But if you took the *average of 50 random households*, and did that over and over, those averages would cluster tightly and symmetrically around the true population mean. Averaging cancels out randomness — that's the CLT at work.

WORKED EXAMPLE

Simulated population: 50,000 household incomes drawn from a right-skewed (exponential-like) distribution, mean $\approx$ Rs. 8,00,000.

Experiment: Draw 1,000 random samples of size $n = 40$ each, compute the mean of each sample, and plot the distribution of those 1,000 sample means.

Result: The original population histogram is heavily skewed (long right tail). The histogram of the 1,000 sample means is close to a symmetric bell curve, centered almost exactly on the true population mean, with a much smaller spread than the original data.

Standard Error: The spread of the sample-means distribution is the **Standard Error (SE)** $= \sigma/\sqrt{n}$ — it shrinks as n grows, which is why larger samples give more precise estimates.

NOTE

This is precisely why we're allowed to build **confidence intervals** and run **t-tests/ANOVA** (Section 1) using formulas that assume normality — CLT guarantees the sample mean behaves normally even when the raw data doesn't.

AI/ML USE CASE

- **Cross-validation:** averaging a model's accuracy across k folds gives a more stable, normally-distributed estimate of true performance than any single train/test split.
- **Bootstrap confidence intervals:** resampling your dataset thousands of times and averaging each time (e.g. to estimate uncertainty around a model metric) relies directly on CLT.
- **Mini-batch gradient descent:** the average gradient computed over a random mini-batch is a noisy-but-unbiased estimate of the true gradient over the full dataset — same averaging logic.

WATCH OUT

CLT is about the distribution of the **sample mean**, not about the raw data becoming normal. Your raw feature can stay skewed forever; only the *average of many samples* approaches normality. Also, CLT needs a reasonably large n and, technically, a finite variance in the population — pathological distributions can need larger n than 30.

TRY IT YOURSELF

In the notebook, change the sample size from $n = 40$ to $n = 5$ and re-run the simulation. Notice the sample-means histogram looks less bell-shaped and more spread out. Then try $n = 200$ and see it get tighter and more normal.

4 Bayes' Theorem Basics**★ KEY IDEA**

Bayes' theorem is a formula for **updating a belief** once new evidence arrives. It combines what you believed *before* (the **prior**) with how well the new evidence fits (the **likelihood**) to give an updated belief (the **posterior**).

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

INTUITION

Read it as: “How likely is A , now that I’ve seen B ?” = “How likely was A before?” $\times$ “How well does B fit with A ?”, all rescaled by “How likely was B overall?”

WORKED EXAMPLE

Real-life example — a medical test. A disease affects **1%** of a population. A test for it is **95% sensitive** (correctly flags 95% of people who truly have it) and **90% specific** (correctly clears 90% of people who don’t have it). You test positive. What’s the actual probability you have the disease?

$$\begin{aligned} P(\text{Disease} | +) &= \frac{P(+ | \text{Disease}) P(\text{Disease})}{P(+ | \text{Disease}) P(\text{Disease}) + P(+ | \text{No Disease}) P(\text{No Disease})} \\ &= \frac{0.95 \times 0.01}{(0.95 \times 0.01) + (0.10 \times 0.99)} = \frac{0.0095}{0.0095 + 0.099} = 0.088 \end{aligned}$$

Interpretation: Even after testing positive, there’s only an **8.8%** chance you actually have the disease! Because the disease is rare (1% prior), most positive results are still false positives from the 99% healthy majority. This is exactly why doctors re-test before treating, and why the **prior** (base rate) matters so much.

AI/ML USE CASE

Naive Bayes spam classifier. Bayes’ theorem is the literal engine behind Naive Bayes classifiers used for spam detection and text classification.

Toy example: Given the word “free” appears in 80% of spam emails but only 5% of genuine (“ham”) emails, and 30% of all incoming email is spam:

$$P(\text{Spam} | \text{“free”}) = \frac{0.80 \times 0.30}{(0.80 \times 0.30) + (0.05 \times 0.70)} = \frac{0.24}{0.24 + 0.035} = 0.873$$

An email containing “free” is now **87.3%** likely to be spam — up from the 30% prior, because the word is strong evidence. Naive Bayes just repeats this update across every word in the email (“naively” assuming words are independent of each other) and multiplies all the evidence together.

NOTE

Bayes' theorem also underlies **Bayesian statistics** more broadly — Bayesian A/B testing, Bayesian optimization for tuning hyperparameters, and Bayesian neural networks all use the same prior $\rightarrow$ likelihood $\rightarrow$ posterior update loop, just with richer distributions instead of simple percentages.

WATCH OUT

The most common real-world mistake is ignoring the **prior** (base rate) and treating the test's own accuracy (95% sensitivity) as if it were the answer. This is called the **base rate fallacy** — always ask “how common was this to begin with?” before trusting a single piece of evidence.

TRY IT YOURSELF

In the notebook, change the disease prevalence from 1% to 20% and recompute $P(\text{Disease} \mid +)$. Notice how much the posterior probability jumps — the rarer the condition, the more a positive result can mislead you.

5 Quick-Reference Summary

Concept	Answers the question	Real-life example	AI/ML use
t-test	Do 2 groups have different means?	Smoker vs non-smoker tips	Compare 2 models/versions
Chi-square	Are 2 categorical variables related?	Class vs Titanic survival	Feature/fairness checks
ANOVA	Do 3+ groups have different means?	Bill size across weekdays	Compare 3+ algorithms
Correlation vs Causation	Does A really cause B?	Ice-cream sales & drowning	Avoid feature-selection traps
CLT & Sampling	Why do sample means behave predictably?	Averaging household incomes	Cross-validation, bootstrap CIs
Bayes' theorem	How do I update belief with evidence?	Disease test result	Naive Bayes spam filter

NOTE**One-line rule of thumb for choosing a hypothesis test:**

Comparing means of **2** groups $\rightarrow$ t-test. Comparing means of **3+** groups $\rightarrow$ ANOVA. Comparing **proportions/categories** $\rightarrow$ chi-square.

End of handbook • See `stats_for_aiml.ipynb` for all runnable code